torch.autograd.Function.vmap#

https://pytorch.org/docs/stable/generated/torch.autograd.Function.vmap.html#

Description:

Define the behavior for this autograd.Function underneath torch.vmap(). For a torch.autograd.Function() to support torch.vmap(), you must either override this static method, or set generate_vmap_rule to True (you may not do both). If you choose to override this staticmethod: it must accept an info object as the first argument. info.batch_size specifies the size of the dimension being vmapped over, while info.randomness is the randomness option passed to torch.vmap(). an in_dims tuple as the second argument. For each arg in args, in_dims has a corresponding Optional[int]. It is None if the arg is not a Tensor or if the arg is not being vmapped over, otherwise, it is an integer specifying what dimension of the Tensor is being vmapped over.

Function Signature

```
static Function.vmap(info, in_dims, *args)[source]#
```

Detailed Documentation

Documentation

Define the behavior for this autograd.Function underneath torch.vmap().

For a torch.autograd.Function() to support torch.vmap(), you must either override this static method, or set generate_vmap_rule to True (you may not do both).

If you choose to override this staticmethod: it must accept

an info object as the first argument. info.batch_size specifies the size of the dimension being vmapped over, while info.randomness is the randomness option passed to torch.vmap().

an in_dims tuple as the second argument. For each arg in args, in_dims has a corresponding Optional[int]. It is None if the arg is not a Tensor or if the arg is not being vmapped over, otherwise, it is an integer specifying what dimension of the Tensor is being vmapped over.

*args, which is the same as the args to forward().

The return of the vmap staticmethod is a tuple of (output, out_dims). Similar to in_dims, out_dims should be of the same structure as output and contain one out_dim per output that specifies if the output has the vmapped dimension and what index it is in.

Please see Extending torch.func with autograd.Function for more details.